Retrieval over the eCFR: Seven Rounds of Measured Experiments

Technical whitepaper — ecfr-mcp retrieval bake-off, 2026-10-12/13;

2026-10-17 addendum: embedding model comparison (bge-small vs bge-base)

Abstract

We built a retrieval layer over a Markdown corpus of the U.S. Code of Federal Regulations (2,278 current-version benefit-program sections) and measured nine configurations against a single frozen 50-query golden set. The simplest production baseline (deterministic citation lookup + dense vectors) beat BM25, hybrid fusion, chunking, and a cross-encoder reranker. Two LLM-assisted techniques then moved the needle: HyDE query expansion (+8pp recall@1) and listwise LLM reranking (+12pp recall@1 over baseline, natural-query recall@3/5 perfect). Total experiment spend: about $0.18. Every number below is measured on the same frozen set — no query was ever edited to make a system pass.

2026-10-17: the shipped index now uses `bge-small-en-v1.5` (384-dim, 127 MB) instead of `bge-base-en-v1.5` (768-dim, 418 MB). The free tier was re-measured for both models on a refreshed corpus snapshot: bge-small 0.60/0.78/0.80 vs bge-base 0.64/0.78/0.84 — a ~4pp r@1 cost for half the RAM. The paid tier was re-measured on bge-small the same day: 0.76/0.88/0.90 ($0.086 for the 50-query run) — r@1 identical to bge-base, the gap

opening at r@3/r@5 where the weaker candidate pool misses outright.

1. Corpus and method

• **Corpus:** `ctrimm/ecfr-markdown` — the eCFR as Markdown, one file per section. Rounds 1–7 ran on the 2026-10-08 snapshot; the 2026-10-17 model comparison on the 2026-10-15 snapshot. Indexed subset: 2,278 current-version sections across benefit programs (SNAP, WIC, TANF, Medicaid/CHIP, LIHEAP, SSI, housing vouchers, Lifeline, child care, UI, school meals, Summer EBT).

• **Embeddings:** shipped index is `BAAI/bge-small-en-v1.5` (384-dim), brute-force cosine, no instruction prefixes. Round 1 (on bge-base) showed bge v1.5 was not trained with `query:` / `passage:` prefixes and they actively hurt (exact-citation queries ranked ~1270th of 2,278, worse than random); the no-prefix finding carries over.

• **Golden set:** 50 queries frozen *before any retrieval ran* — 5 exact citations ("7 CFR 273.9"), 33 natural questions, 12 paraphrases, stratified across 14 program areas. Sealed: queries are added, never edited to fit the system.

• **Metric:** recall@1 / @3 / @5, overall and per query type.

2. The production baseline

A query matching `\d+ CFR [\w.]+` is a **deterministic lookup**, not retrieval — a query that IS a citation resolves by key (recall 1.00 by construction, and correct by design). Everything else ranks by cosine similarity over whole-section vectors. BM25 (SQLite FTS5, title 3x

weight) was built and kept as a fallback/debug leg, but RRF fusion never beat vectors+citation on any k, so BM25 is not in the ranking path.

3. Seven rounds (chart: progression.png, recall_all.png)

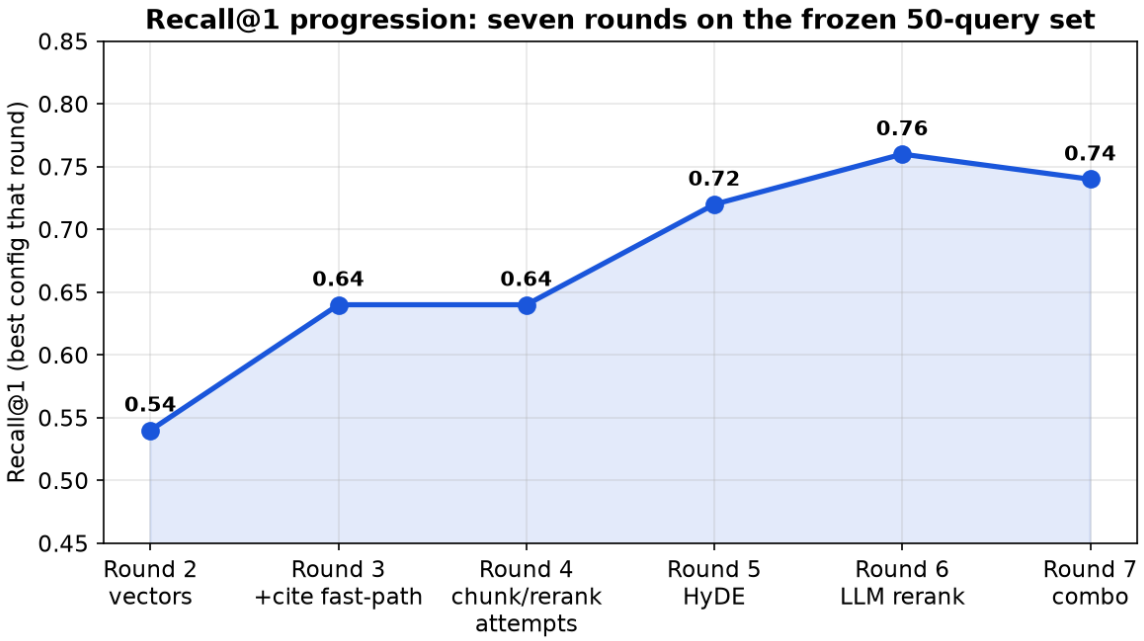


Figure 1 — Recall@1 progression across the seven rounds.

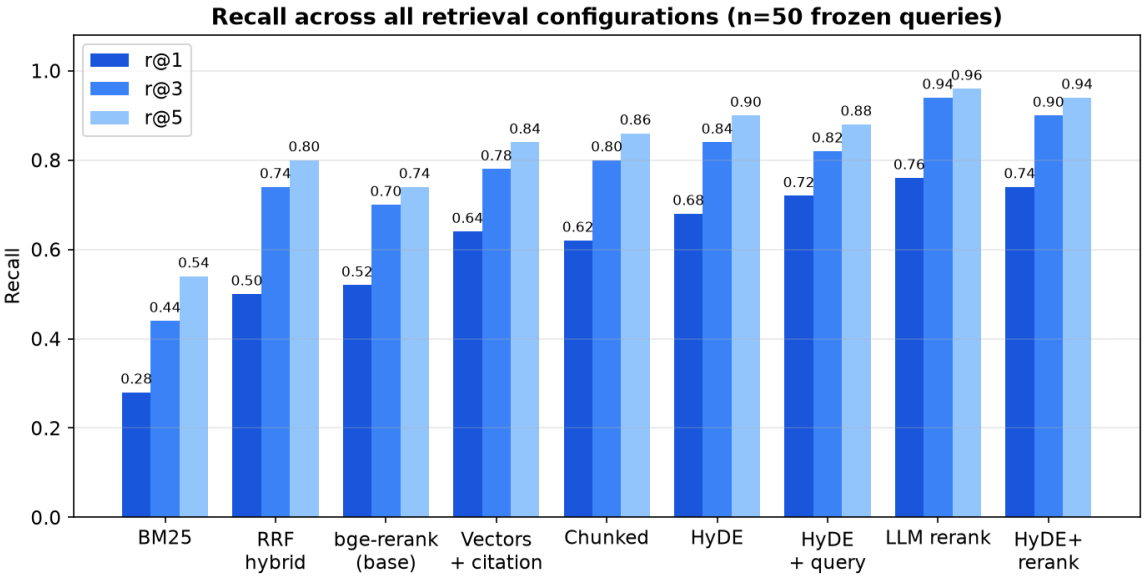


Figure 2 — Recall across all nine configurations.

All rounds below ran on bge-base (768-dim), corpus snapshot 2026-10-08:

Round	System	r@1	r@3	r@5	Cost

2	Vectors, no prefixes	0.540	0.720	0.800	$0
3a	BM25 alone	0.280	0.440	0.540	$0
3b	RRF hybrid (best k)	0.500	0.760	0.840	$0
3c	**Vectors + citation fast-path**	**0.640**	**0.780**	**0.840**	$0
4a	Chunked index (6,222 chunks, best-chunk)	0.620	0.800	0.860	$0
4b	bge-reranker-base, sections	0.520	0.700	0.740	$0
4c	bge-reranker-base, chunks	0.520	0.700	0.760	$0
5a	HyDE passage only	0.680	0.840	0.900	$0.006
5b	**HyDE + query averaged**	**0.720**	**0.820**	**0.880**	$0.006
6	**LLM rerank, query candidates**	**0.760**	**0.940**	**0.960**	$0.09
7	HyDE candidates + LLM rerank	0.740	0.900	0.940	$0.18

Per-type recall@1 for the three leaders (chart: recall_by_type.png):

System	natural (n=33)	paraphrase (n=12)	exact (n=5)
Vectors + citation	0.667	0.417	1.000
HyDE + query	0.758	0.500	1.000
LLM rerank	0.788	0.583	1.000

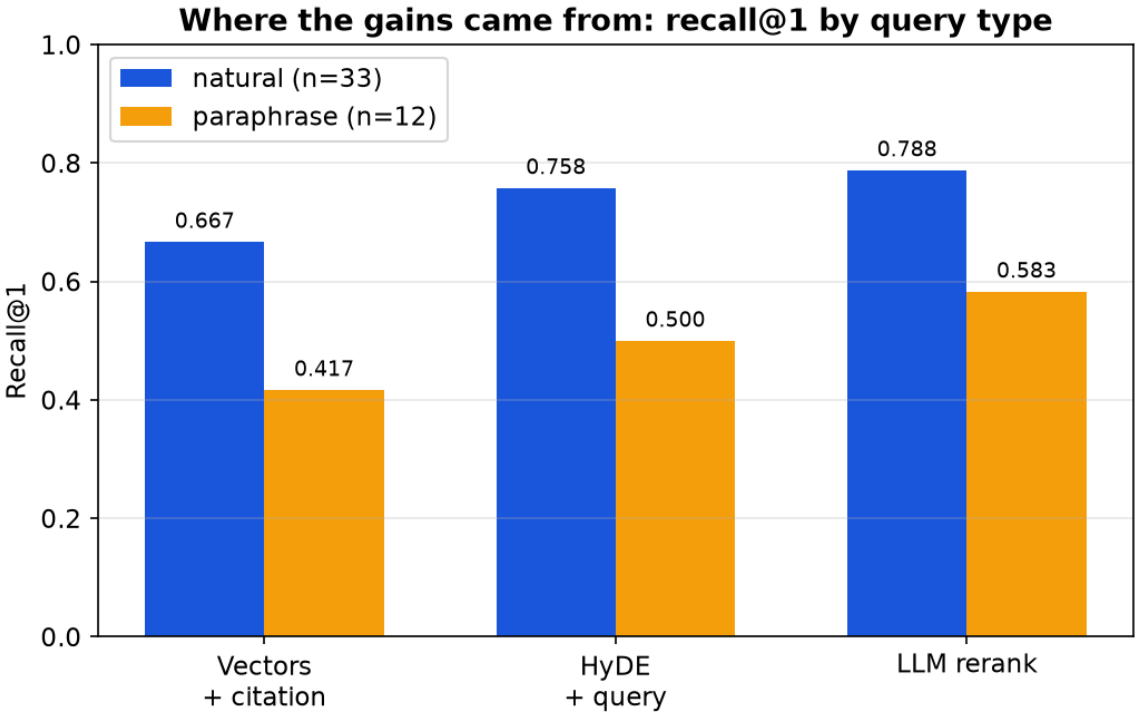


Figure 3 — Recall@1 by query type for the three leaders.

LLM rerank additionally reaches natural r@3/r@5 = **1.000** (perfect).

3b. 2026-10-17: bge-small vs bge-base (free tier)

Same frozen 50 queries, same code, refreshed corpus (2026-10-15); only the embedding model differs. Full detail in `golden/results-2026-10-17.md`.

model	r@1	r@3	r@5
bge-base (768-dim, rounds 1–7)	**0.64**	**0.78**	**0.84**
bge-small (384-dim, shipped)	0.60	**0.78**	0.80

By query type, r@1 / r@3 / r@5:

model	exact (n=5)	natural (n=33)	paraphrase (n=12)
bge-base	1.00 / 1.00 / 1.00	0.667 / 0.788 / 0.848	0.417 / 0.667 / 0.750
bge-small	1.00 / 1.00 / 1.00	0.606 / 0.788 / 0.818	0.417 / 0.667 / 0.667

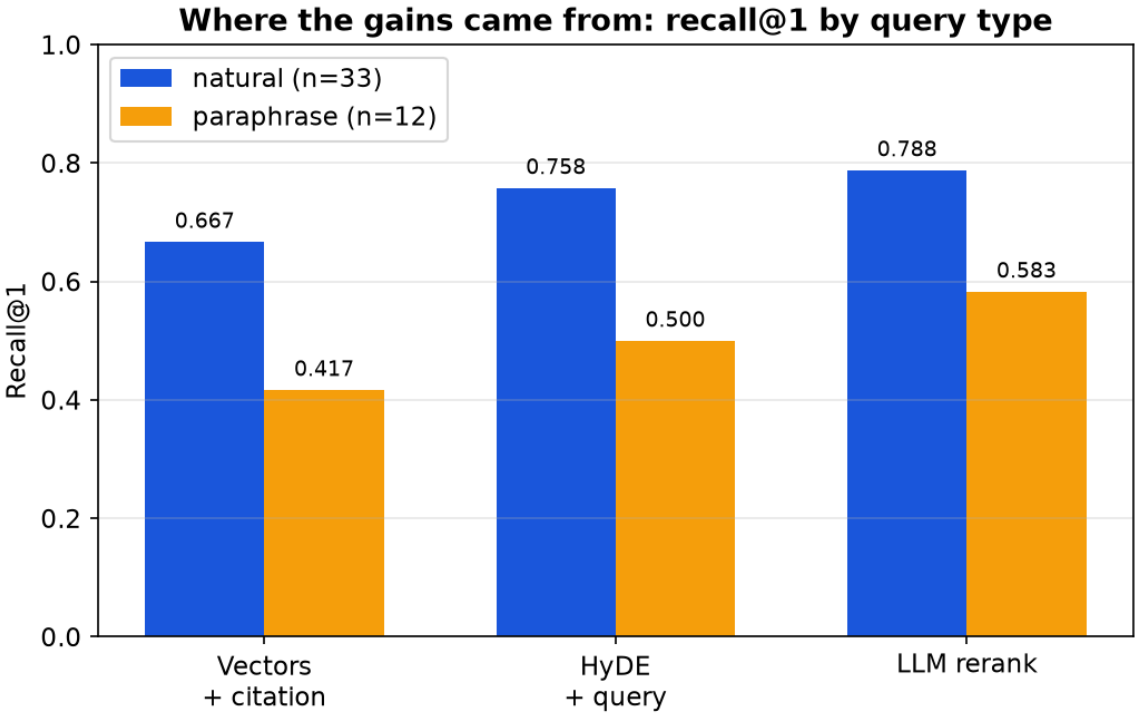


Figure 3 — Recall@1 by query type for the three leaders.

bge-base reproduces its 2026-10-12 figures exactly on the refreshed corpus, so the eval is stable across the snapshot refresh and the gap above is the model. The switch costs 2 queries at r@1 (natural 0.667 → 0.606) and 2 at r@5 — within noise at n=50, but pointing the same way in every slice where the two differ. bge-small was chosen for size: 127 MB vs 418 MB download, about half the RAM. Whether ~4pp of r@1 is worth it is a product call; reverting is a rebuild of `data/` with `--model BAAI/bge-base-en-v1.5`.

The paid tier (LLM rerank over chunk candidates) was re-measured on bge-small 2026-10-17: 0.76/0.88/0.90 ($0.086; scores in `golden/llm_rerank_scores_bgesmall.jsonl`). r@1 matches bge-base exactly; the ~6pp gap at r@3/r@5 is the candidate pool, not the reranker.

4. Findings

1. **Prefixes hurt.** bge-base-en-v1.5 was not trained with `query:` / `passage:` prefixes; both sides landed out-of-distribution. Removing them: 0.40 -> 0.54 r@1.

2. **A citation is a lookup.** Short citation strings carry almost no semantic signal; dense retrieval ranked "20 CFR 416.1100" 592nd. The deterministic fast-path fixes the whole category.

3. **BM25-first loses.** RRF fusion (k=5..60) never beats vectors + citation on any cutoff. BM25 stays as fallback only.

4. **39.7% of sections exceed bge's 512-token limit** and were silently truncated — real data loss, but chunking (450-token windows, 64 overlap, title-prepended, section = best chunk) measured as a wash (0.62/0.80/0.86): it fixed truncation in principle while diluting topical signal on paraphrase queries (r@1 0.417 -> 0.250).

5. **bge-reranker-base hurts in both configurations** (0.52/0.70/0.74 on sections, 0.52/0.70/0.76 on chunks). Reranking chunks eliminated the truncation confound, so the cause is model/domain mismatch — MS MARCO training rewards surface lexical overlap and promotes distractor docs (e.g. the OMB control-number list that cites everything).

6. **HyDE is the first real win** (+8pp r@1, $0.0056 total): a flash model writes a 4-6 sentence CFR-style passage answering the query; embedding the passage (alone, or averaged with the query vector) beats the query vector alone. HyDE-only is strongest on paraphrase (0.750); the average is the balanced best.

7. **Listwise LLM reranking wins overall** (gemini-2.5-flash-lite scores top-50 chunks 10-at-a-time, section = best chunk): 0.76/0.88/0.90 on bge-small for $0.086 (2026-10-17; 0.76/0.94/0.96 on bge-base, 2026-10-13). Operational note: deepseek-v4-flash via SiliconFlow returned intermittent null-content HTTP 200s in streaks and was abandoned after gemini went 5/5 clean on identical prompts.
8. **The combo does not stack.** HyDE candidates + LLM rerank scored 0.74/0.90/0.94 — worse everywhere than query candidates + rerank. The reranker does the heavy lifting; once it sees the top-50, the candidate source barely matters. Rejected (extra LLM call, no gain).
9. **Smaller embeddings cost ~4pp r@1** (2026-10-17). bge-small-en-v1.5 (384-dim) trails bge-base 0.60 vs 0.64 r@1 on the same frozen set, identical at r@3. Shipped anyway for the size win (127 MB vs 418 MB, ~half the RAM) — a product tradeoff, revertible with one rebuild.

5. Cost analysis (chart: cost_frontier.png)

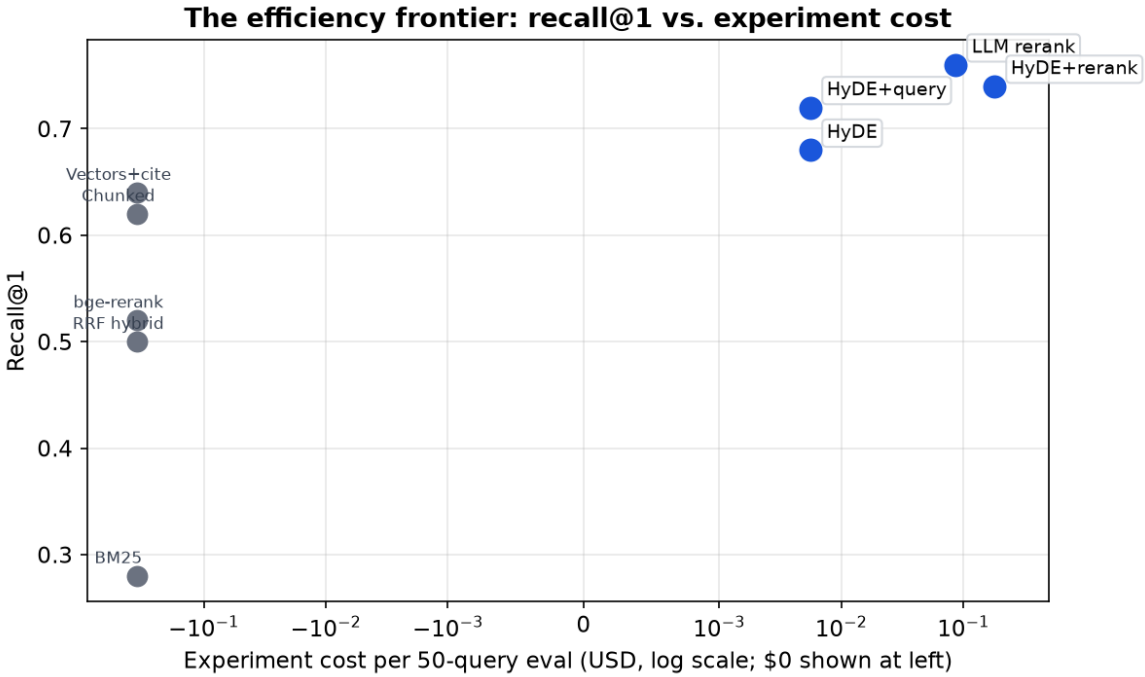


Figure 4 — The efficiency frontier: recall@1 vs experiment cost.

The efficiency frontier is steep and short: $0.18 of total experiment spend bought +12pp r@1 / +16pp r@3 over the free baseline. Per-query serving costs at the winning config: ~6 flash-model calls, roughly $0.002 and 30-60s sequential latency (batchable). Whether that belongs in the default serving path is a product decision — see deployment tiers below.

6. Deployment: env-var-gated tiers (MCP server)

The server starts in the tier its environment supports — no flags, no config files:

• **Free tier** (no key): citation fast-path + local vectors (bge-small as shipped). 0.60/0.78/0.80, fully offline after install, $0/query.

• **Best tier** (`OPENROUTER_API_KEY` set in the user's shell): adds the listwise LLM rerank over top-50 chunks. 0.76/0.88/0.90 on the shipped bge-small index (2026-10-17). ~$0.002/query.

• **Override:** `ECFR_MCP_TIER=free|best` pins a tier regardless of the key (e.g. force free for latency-sensitive use, or test best without exporting a key globally).

Degradation is graceful: if the key is absent or the LLM call fails, the server serves the free tier rather than erroring. The golden set and all experiment scripts ship in the repo so any tier choice stays re-measurable.

7. Limitations

n=50 is small — differences under ~0.04 are noise, and several "wins" should be re-confirmed on a larger set before hardening into lore. All embedding work ran CPU-only on a 7GB box (bge-reranker-v2-m3 OOM-killed there; the LLM judge replaced it). Two corpus snapshots (eCFR 2026-10-08 for rounds 1–7, 2026-10-15 for the model comparison); version-chaining retrieval is built but not evaluated here.

8. Reproducibility

Repo: `ctrimm/ecfr-mcp` (private). `golden/eval.json` (frozen set), `golden/eval.py`, `golden/eval_chunks.py`, `golden/exp_rerank.py`, `golden/exp_rerank_chunks.py`, `golden/exp_hyde.py`, `golden/exp_llm_rerank.py`, `golden/eval_server.py` (free-tier numbers via the server's own code); `golden/results-2026-10-12.md` (rounds 1–7,

bge-base), `golden/results-2026-10-17.md` (bge-small vs bge-base);

`DECISION_LOG.md` (each data-backed decision with evidence and date).